

P I S A L P R A J W A L

CONTACT INFO

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ABOUT ME

A dynamic researcher with a multidisciplinary background, specializing in materials science and machine learning. My current focus involves cutting-edge research in machine learning workflows for efficient CO₂ conversion, aiming to drive sustainable solutions for environmental challenges. Committed to advancing clean energy and catalysis, I aspire to make a significant impact on global sustainability in the fight against climate change.

EDUCATION

DOCTORAL RESEARCHER Jan 2025 - Ongoing
Technical University of Munich | Garching bei München, Germany

- Continuing doctoral studies in AI-guided CO₂ conversion
- Key concepts: Machine learning, Simulations, High performance computing

DOCTORAL RESEARCHER Aug 2022 - Dec 2024
Aalto University | Helsinki, Finland

- Doctoral studies under Prof. Patrick Rinke on AI-guided CO₂ conversion
- Key concepts: Project AICon, Open Catalyst Project, Catalysis, Machine Learning

BS-MS DUAL DEGREE IN CHEMISTRY Aug 2017 - July 2022
Indian Institute of Science Education and Research | Tirupati, India

- Cumulative Grade Point Average: 8.4 out of 10

Higher Secondary Certificate until 2017
Fergusson College | Pune, India

- Percentage marks obtained: 90.7%

EXPERIENCE

VISITING RESEARCHER May 2023
Carnegie Mellon University | Pittsburgh, USA

- Worked on applying Open Catalyst Project models to my Doctoral Project

RESEARCH INTERN **June 2021- May 2022**
CSIR - National Chemical Laboratory | Pune, India
(Master's Thesis)

- Computational modeling of catalysts based on nanomaterials for CO₂ conversion

PROJECT INTERN **Jan 2019- April 2019**
Indian Institute of Science Education and Research | Tirupati, India
(Semester Project)

- Studying pi-pi stacking interactions in a benzene dimer system using Density Functional Theory

SUMMER INTERN **April 2018 - July 2018**
National Centre for Microbial Resource | Pune, India
(Summer Project)

- Bacterial Identification from river-waters using genome sequencing and protein mass spectrometry

ACHIEVEMENTS

- | | |
|---------------------|---|
| Awards | <ul style="list-style-type: none">• Honorary mention for the poster presentation at Physics Days 2023, Tampere University• Student of the year award from LIC India for excellent academic performance during the academic year 2017 |
| Scholarships | <ul style="list-style-type: none">• Innovation in Science Pursuit for Inspired Research (INSPIRE) from DST, Govt. of India for top 1% students graduating with higher secondary certificate (2017-2022) |

PUBLICATIONS

- | | |
|-----------------|---|
| Articles | Pisal Prajwal, Krejčí Ondřej, and Rinke Patrick (July 2025). "Machine learning accelerated descriptor design for catalyst discovery in CO ₂ to methanol conversion". en. In: <i>npj Computational Materials</i> 11. DOI: 10.1038/s41524-025-01664-9. |
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CONFERENCES

- | | |
|---|-----------------------|
| Psi-k Conference 2025
<i>EPFL Lausanne</i> Lausanne, Switzerland
(Contributed Talk on PhD work) | August, 2025 |
| Machine Learning for Materials Discovery (ML4MD) 2025
<i>Aalto University</i> Espoo, Finland
(Poster Presentation on PhD work) | May, 2025 |
| DPG Spring Meeting 2024
<i>Technische Universität Berlin</i> Berlin
(Oral Presentation on PhD work) | March, 2024 |
| International conference for Carbon Dioxide Utilization (ICCDUXX)
<i>University of Bari</i> Bari, Italy
(Oral Presentation on AI-guided CO ₂ conversion workflow development) | June, 2023 |
| Actively Learning Materials Science (AL4MS)
<i>Aalto University</i> Espoo, Finland
(Poster presentation on machine learning model for CO ₂ catalyst data) | February, 2023 |

Physics Days 2023

Tampere University | Tampere, Finland

(Poster presentation on machine learning model for CO₂ catalyst data)

March, 2023

Nanoscience Days 2022

University of Jyväskylä | Jyväskylä, Finland

(Poster presentation on Masters Thesis work)

August, 2022

SKILLS

Programming

Core expertise

Python, Linux, High-performance Computing, Git

Materials modeling, Quantum Chemistry Simulations, Machine Learning

Languages

- Marathi (Native)
- English (Full professional proficiency)
- Hindi (Full professional proficiency)
- German (Limited working proficiency)
- Finnish (Basic)
- Kannada (Basic)

REFERENCES

Prof. Dr. Patrick Rinke

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Department of Physics
Technical University of Munich
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Prof. Dr. Helge Stein

Professor
Department of Chemsitry
Technical University of Munich
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